PlotlyBackend()

⚠ Failed to load integration with PlotlyBase & PlotlyKaleido.
exception:

Error message from Main

The package PlotlyBase.jl could not load because it failed to initialize.

That's not nice! Things you could try:

- Restart the notebook.
- Try a different Julia version.
- Contact the developers of PlotlyBase.jl about this error.

You might find useful information in the package installation log:

```
Precompiling Plots...
   3686.7 ms  ✓ ColorVectorSpace → SpecialFunctionsExt
   6302.9 ms  ✓ Unitful → ForwardDiffExt
  2 dependencies successfully precompiled in 8 seconds. 198 already precomp
iled.
Precompiling SparseArraysExt...
   4471.7 ms  ✓ Latexify → SparseArraysExt
  1 dependency successfully precompiled in 5 seconds. 24 already precompile
d.
Precompiling Integrals...
   6927.2 ms  ✓ HCubature
  10025.1 ms  ✓ Distributions → DistributionsTestExt
   1039.6 ms  ✓ MonteCarloIntegration
```

Sections:

Units:

The Problem PMO

The Solution

Raman Mode
IR Mode
Potential

Fourier Transfroms ☐

Raman Mode
IR Mode
Potential

Short-time FT ☐

Raman Mode
IR Mode
Potential

Wavelet Transforms ☐

Raman Mode
IR Mode
Potential

Units:

```
begin
    amu= 1.6605402e-27  # [kg]
    eV = 1.60217733e-19  # [J]
    Å= 1.0e-10  # [m]
    THz = 1.0e12  # [s⁻¹]
    cm⁻¹= 1e2 # [m⁻¹]
    ps= 1e-12 #[s]
    phy=1e-3*eV/(amu*Å^2)
    T=sqrt(1e-3eV/amu)/(Å*THz)
    println("amu= $amu kg")
    println("eV= $eV J")
    println("Å= $Å m")
    println("THz= $THz s⁻¹")
    println("cm⁻¹= $cm⁻¹ m⁻¹")
    println("ps= $ps s")
    println("T= $T THz")
    println("phy= $phy meV*amu⁻¹*Å⁻²")
end
```

```
amu= 1.6605402e-27 kg
eV= 1.60217733e-19 J
Å= 1.0e-10 m
THz= 1.0e12 s⁻¹
cm⁻¹= 100.0 m⁻¹
ps= 1.0e-12 s
T= 3.106208431750202 THz
phy= 9.648530821476047e24 meV*amu⁻¹*Å⁻²
```

The Problem PMO

We now look at a potential given by:

$$V(Q_1, Q_2) = \frac{\Omega_1^2 Q_1^2}{2} + \frac{\Omega_2^2 Q_2^2}{2} + \frac{a_3 Q_1^3}{3} + \frac{b_4 Q_2^4}{4} - \frac{g Q_1 Q_2^2}{2}$$

The equation of motion for Q_1 and Q_2 are, thus, given by:

$$\ddot{Q}_1 = -\frac{\partial V}{\partial Q_1} = -\Omega_1^2 Q_1 - a_3 Q_1^2 + \frac{g Q_2^2}{2} - 5\% \Omega_1 \dot{Q}_1$$

$$\ddot{Q}_2 = -\frac{\partial V}{\partial Q_2} = -\Omega_2^2 Q_2 - b_4 Q_2^3 + g Q_1 Q_2 + F(t) - 5\% \Omega_2 \dot{Q}_2$$

Where F(t) is the driving force, which we will take to be $F(t) = f * e^{-\frac{t^2}{2\sigma^2}} \sin(\omega_p t)$

The Solution

Damping: ☑

f = 500.0 Submit meV×Å^{-1}×(√amu)$^{-1}$

	timestamp	value1	value2	value3	value4
1	-1.75	0.1	0.1	0.0	0.0
2	-1.74909	0.0999965	0.0999417	-0.00773277	-0.128197
3	-1.74822	0.0999865	0.0997765	-0.0151465	-0.250813
4	-1.74706	0.0999633	0.0993929	-0.0249718	-0.412644
5	-1.74581	0.0999255	0.0987691	-0.0355779	-0.586156
6	-1.74437	0.0998654	0.0977814	-0.0478105	-0.784276
7	-1.74281	0.0997803	0.0963886	-0.0610778	-0.996088
8	-1.7411	0.0996634	0.0944897	-0.0755762	-1.22308
9	-1.73927	0.0995109	0.092034	-0.0910709	-1.45948
10	-1.73732	0.0993172	0.0889482	-0.107567	-1.70295

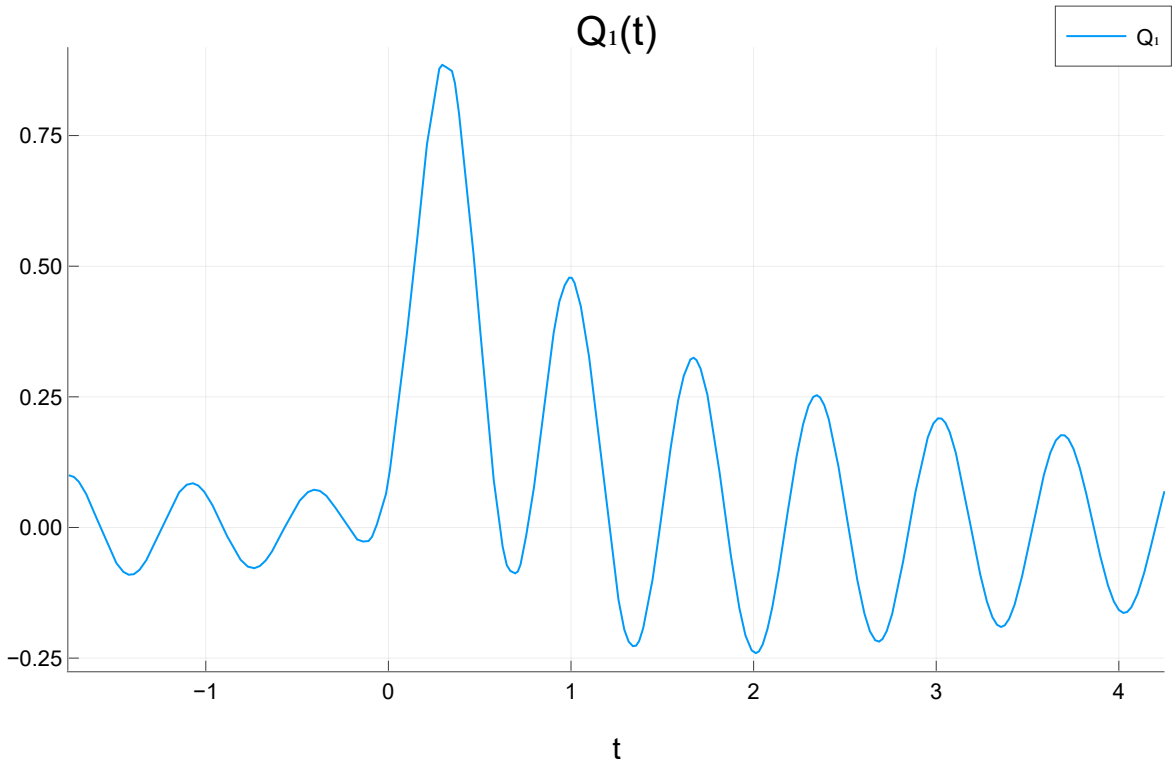
⋮ more

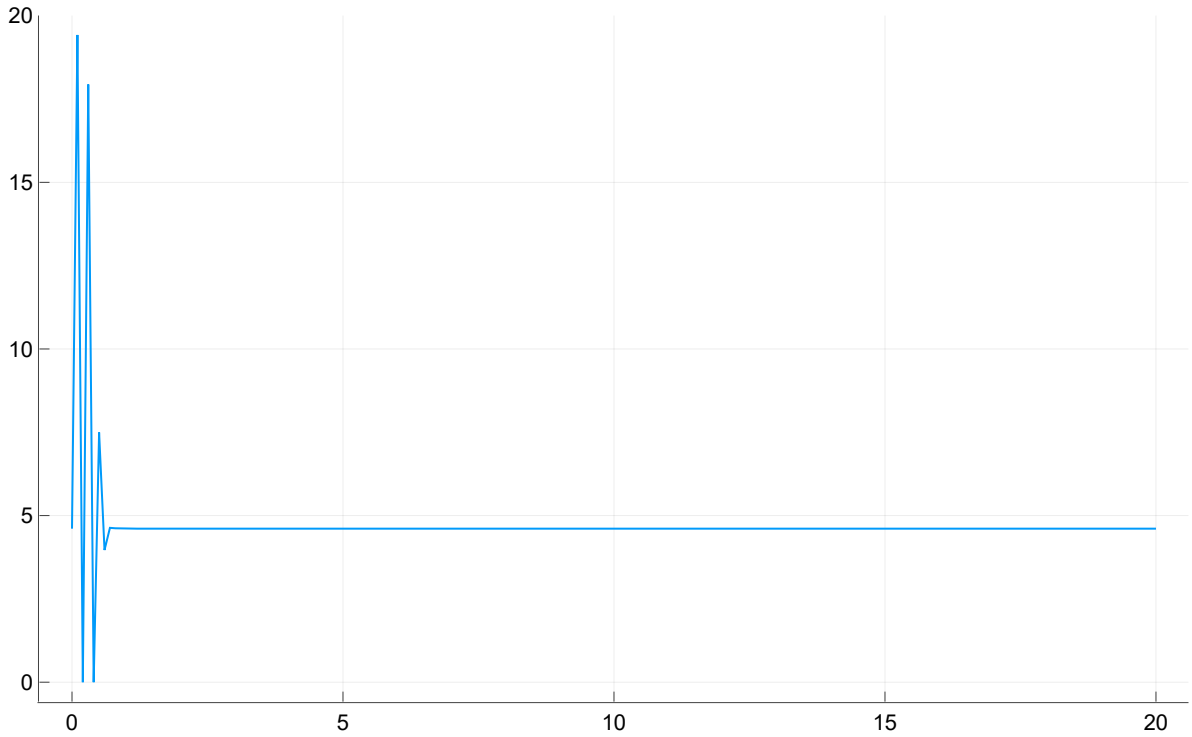
```
begin
    if prob=="PMO"
        Q₁ = 0.1 #Å√amu
        Q₂ = 0.1 #Å√amu
        dQ₁ = 0.0 #Å√(amu)s⁻¹
        dQ₂ = 0.0 #Å√(amu)s⁻¹
        Ω₁ = sqrt(87.09) #phy)
        Ω₂ = sqrt(1416.13) #phy)
        g = 51.74 #phy/(sqrt(amu)*Å)
        a₃ = 5.82 #phy/(sqrt(amu)*Å)
        b₄ = 80.24 #phy/(amu*Å^2)
        σ = 0.2 #ps
        ωₚ = 0.95*Ω₂

        dur=(-1.75,4.25)

        F(t)=f*exp.(-t.^2/2σ^2).*sin.(ωₚ*t)

        function V(x,y,t)
            Ω₁^2*x.^2/2+Ω₂^2*y.^2/2+a₃.*x.^3/3+b₄*y.^4/4-g*x.*y.^2/2
        end

        Q=[Q₁,Q₂,dQ₁,dQ₂]

        function motion(du,u,p,t)
            du[1]=u[3]
            du[2]=u[4]
            du[3]=-Ω₁^2*u[1]-a₃*u[1]^2+g*u[2]^2/2-d*0.05*Ω₁*u[3]
            du[4]=-Ω₂^2*u[2]-b₄*u[2]^3+g*u[1]*u[2]+p(t)-d*0.05*Ω₂*u[4]
        end

        Sol=solve(ODEProblem(motion,Q,dur,F),AutoTsit5(Rosenbrock23()),abstol=1e-8,reltol=1e-8)

    elseif prob=="LCO"
```

```
        #for beating use [0.0,0.0,±0.01,±0.01] or [0.0,0.0,±0.01,∓0.01] for weird signal [0.0,0.0,0.1,-0.1] or familiar signal [0.0,0.0,0.1,x>29] all with no damping

        Q₁ = 0.01 #Å√amu
        Q₂ = 0.1 #Å√amu
        dQ₁ = 0.0 #Å√(amu)s⁻¹
        dQ₂ = 0.0 #Å√(amu)s⁻¹
        Ω₁ = sqrt(103.55) #phy)
        Ω₂ = sqrt(1462.33) #phy)
        g = 46.98 #phy/(amu*Å^2)
        a₄ = 8.36 #phy/(amu*Å^2)
        b₄ = 135.18 #phy/(amu*Å^2)
        σ = 0.2 #ps
        ωₚ = 0.95*Ω₂

        dur=(-1,3)

        F(t)=f*exp.(-t.^2/2σ^2).*sin.(ωₚ*t)

        function V(x,y,t)
            Ω₁^2*x.^2/2+Ω₂^2*y.^2/2+a₄.*x.^4/4+b₄*y.^4/4-g*x.^2 .*y.^2/2
        end

        Q=[Q₁,Q₂,dQ₁,dQ₂]

        function motion(du,u,p,t)
            du[1]=u[3]
            du[2]=u[4]
            du[3]=-Ω₁^2*u[1]-a₄*u[1]^3+g*u[1]*u[2]^2-d*0.05*Ω₁*u[3]
            du[4]=-Ω₂^2*u[2]-b₄*u[2]^3+g*u[1]^2*u[2]+p(t)-d*0.05*Ω₂*u[4]
        end

        Sol=solve(ODEProblem(motion,Q,dur,F),AutoTsit5(Rosenbrock23()),abstol=1e-8,reltol=1e-8)

    elseif prob=="PMO Inverse"

        Q₁ = 0.1 #Å√amu
        Q₂ = 0.1 #Å√amu
        dQ₁ = 0.0 #Å√(amu)s⁻¹
        dQ₂ = 0.0 #Å√(amu)s⁻¹
        Ω₁ = 2.09977*2π/T #phy)
        Ω₂ = 2.37265*2π/T #phy)
        g = 51.74 #phy/(sqrt(amu)*Å)
        a₃ = 5.82 #phy/(sqrt(amu)*Å)
        b₄ = 80.24 #phy/(amu*Å^2)
        σ = 0.2 #ps
        ωₚ = 0.95*Ω₂

        dur=(-1,20)

        F(t)=f*exp.(-t.^2/2σ^2).*sin.(ωₚ*t)

        function V(x,y,t)
            Ω₁^2*x.^2/2+Ω₂^2*y.^2/2+a₃.*x.^3/3+b₄*y.^4/4-g*x.*y.^2/2
        end
```

```
        Q=[Q₁,Q₂,dQ₁,dQ₂]

        function motion(du,u,p,t)
            du[1]=u[3]
            du[2]=u[4]
            du[3]=-Ω₁^2*u[1]-a₃*u[1]^2+g*u[1]*u[2]-d*0.05*Ω₁*u[3]
            du[4]=-Ω₂^2*u[2]-b₄*u[2]^3+g*u[1]^2/2+p(t)-d*0.05*Ω₂*u[4]
        end

        Sol=solve(ODEProblem(motion,Q,dur,F),AutoTsit5(Rosenbrock23()),abstol=1e-8,reltol=1e-8)

    elseif prob=="Chirp"
        Q₁ = 0.1 #Å√amu
        Q₂ = 0.01 #Å√amu
        dQ₁ = 0.0 #Å√(amu)s⁻¹
        dQ₂ = 0.0 #Å√(amu)s⁻¹
        Ω₁ = 2.09977*2π/T #2.09977*2π/T #phy)
        Ω₂ = 2.37365*2π/T #2.37365*2π/T #phy)
        g₁ = ((4-0.1^2)/4-(2*2π/T)^2/Ω₁^2)*Ω₁^2 #phy/(sqrt(amu)*Å)
        g₂ = (4*0.5*Ω₂)/(0.1*2*Ω₂+1) #0.1*Ω₂ #phy/(sqrt(amu)*Å)
        a₄ = 0 #0.05*Ω₁ #phy/(sqrt(amu)*Å)
        b₄ = 0 #0.05*Ω₂ #phy/(amu*Å^2)
        σ = 5 #ps

        dur=(-1,20)

        F(t)=0.5*(sign.(t).+1).*(f*exp.(-t.^2/2σ^2))

        function V(x,y,t)
            (Ω₁^2 .-g₁.*F(t)).*x.^2/2+Ω₂^2*y.^2/2+a₄.*x.^4/4+b₄*y.^4/4 .-g₂.*y.*F(t)
        end

        Q=[Q₁,Q₂,dQ₁,dQ₂]

        function motion(du,u,p,t)
            du[1]=u[3]
            du[2]=u[4]
            du[3]=-Ω₁^2*u[1]-a₄*u[1]^3+g₁*u[1]*p(t)-d*0.1*Ω₁*u[3]
            du[4]=-Ω₂^2*u[2]-b₄*u[2]^3+g₂*p(t)-d*0.1*Ω₂*u[4]
        end

        Sol=solve(ODEProblem(motion,Q,dur,F),AutoTsit5(Rosenbrock23()),abstol=1e-8,reltol=1e-8)
    end
```

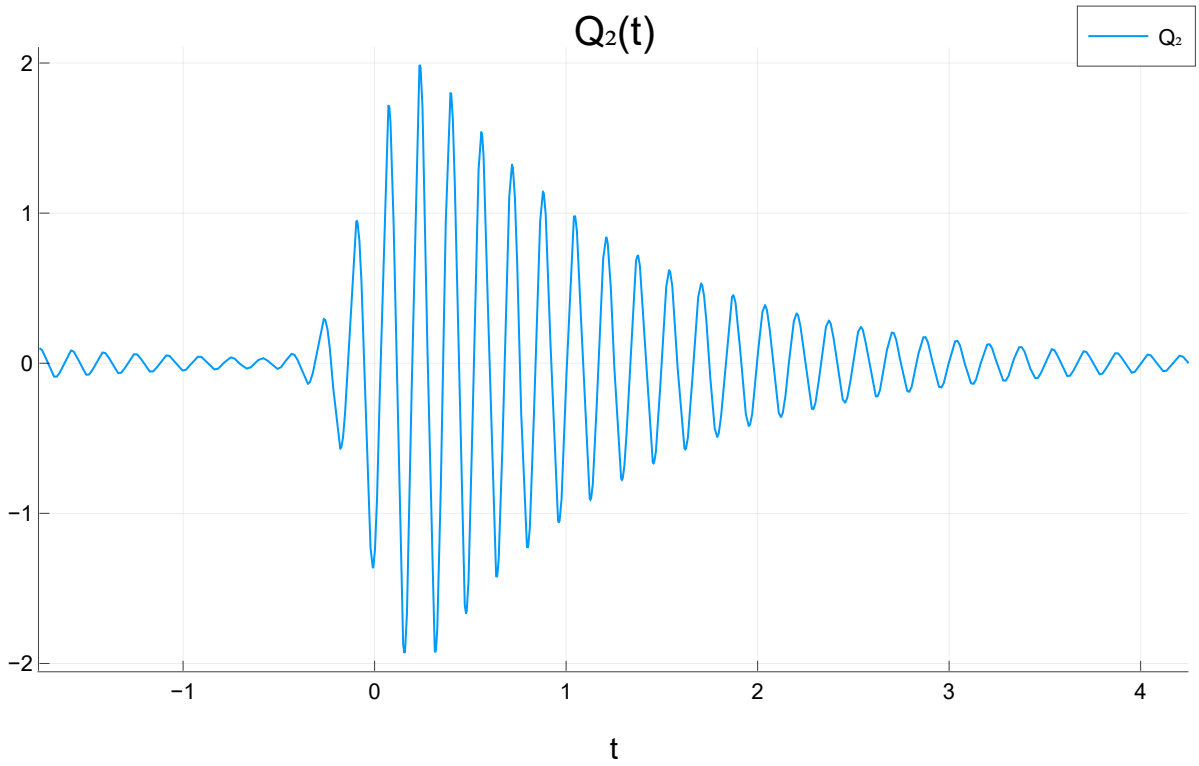
Raman Mode



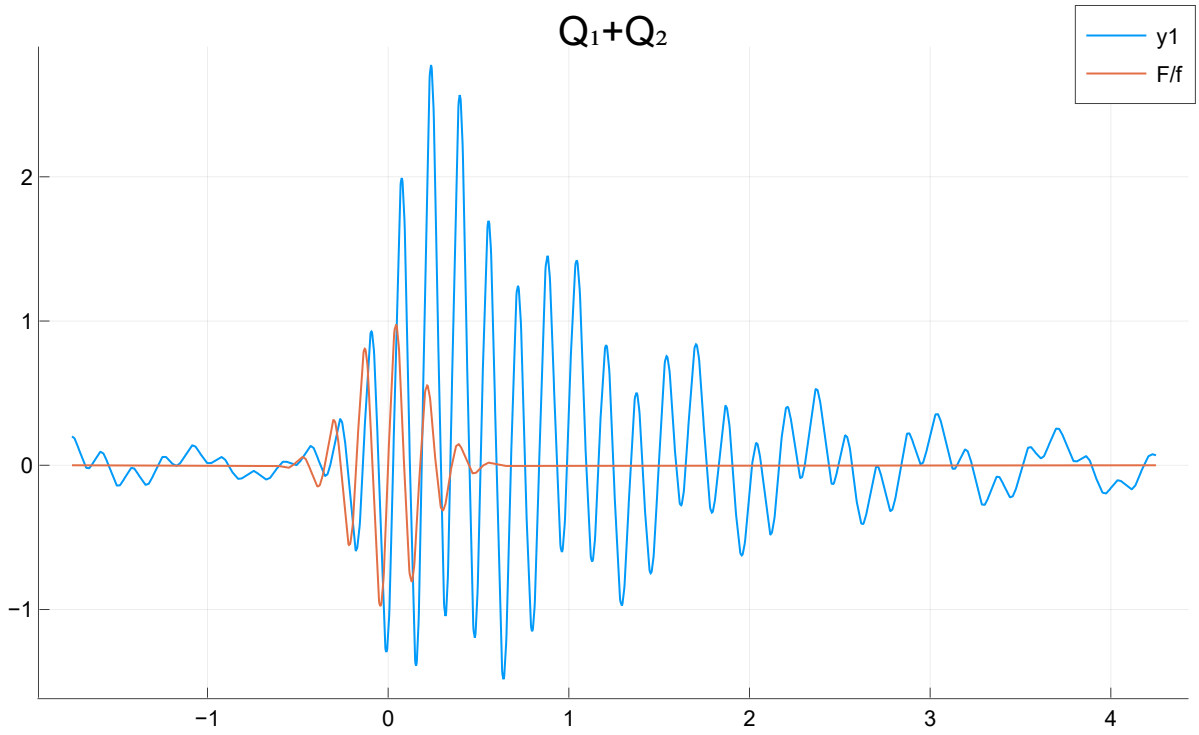



```
⟨Q₁⟩=0.07462233409731549
```

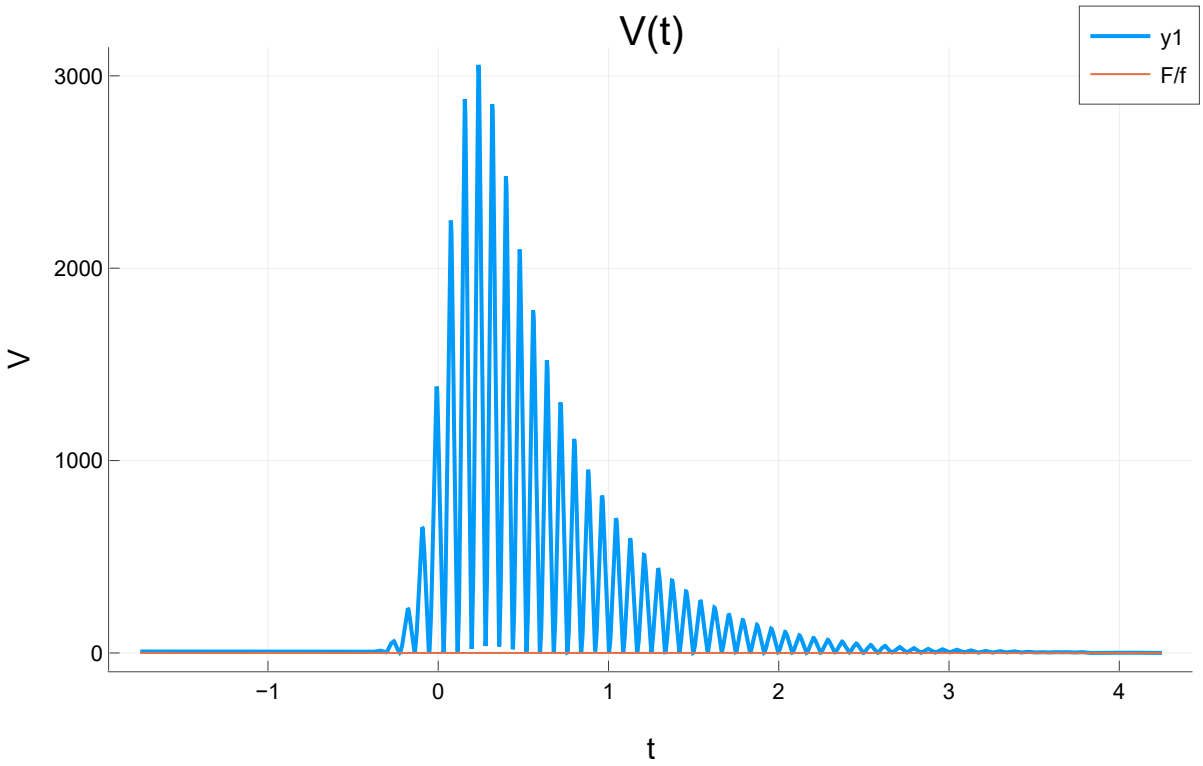
IR Mode




```
⟨Q₂⟩=0.0002368207155100411
```



Potential

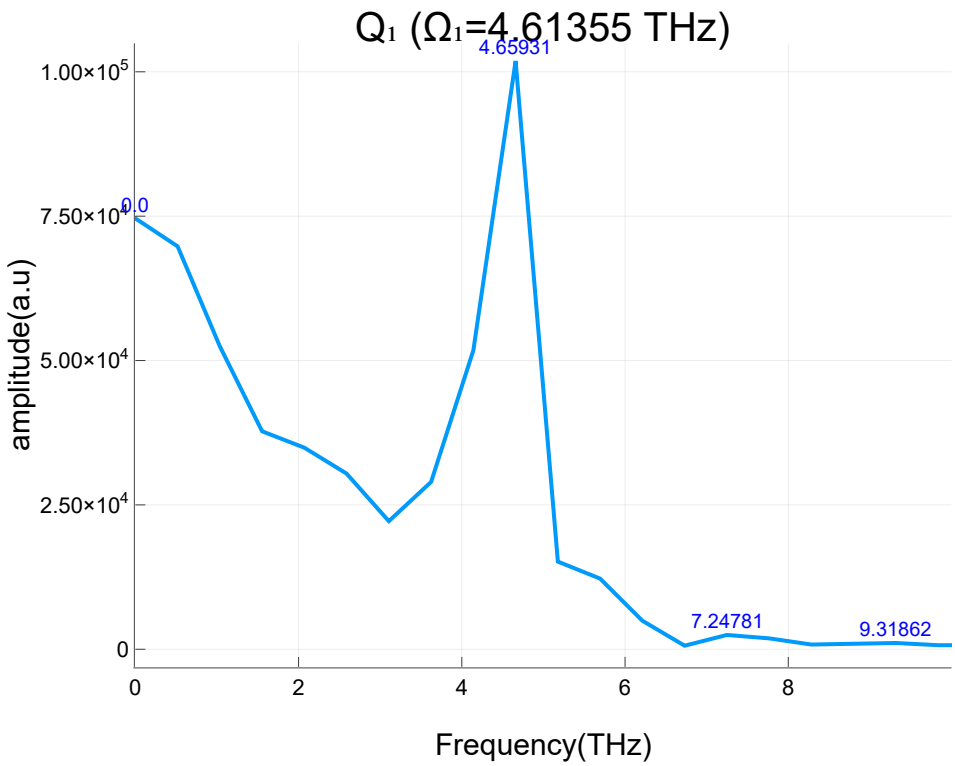



```
⟨V⟩=202.4697198622283
```

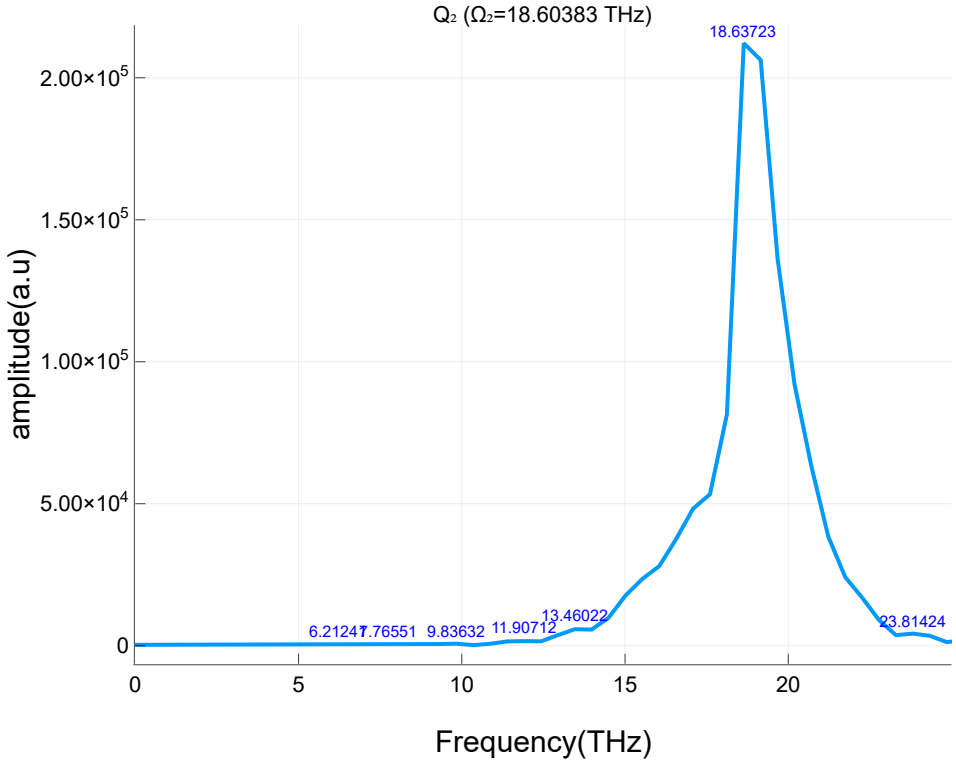
Fourier Transfroms

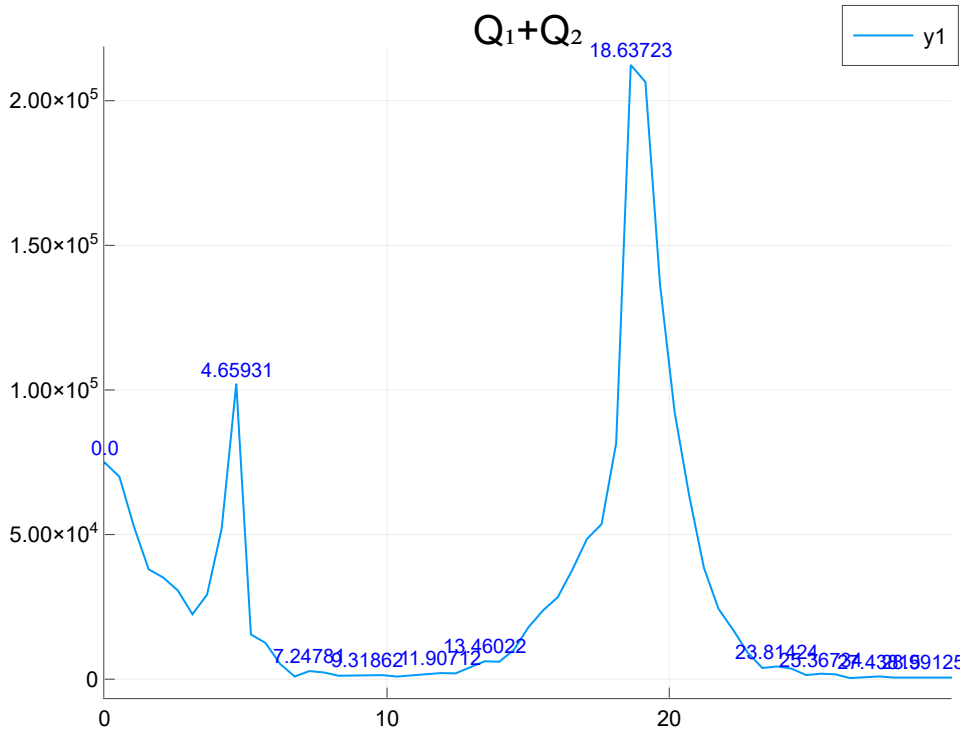
```
[[0.0153092, 0.0153092, 0.0153092, 0.0153092, 0.0153092, 0.0153092, 0.0153092, 0.0153092,
```

Raman Mode

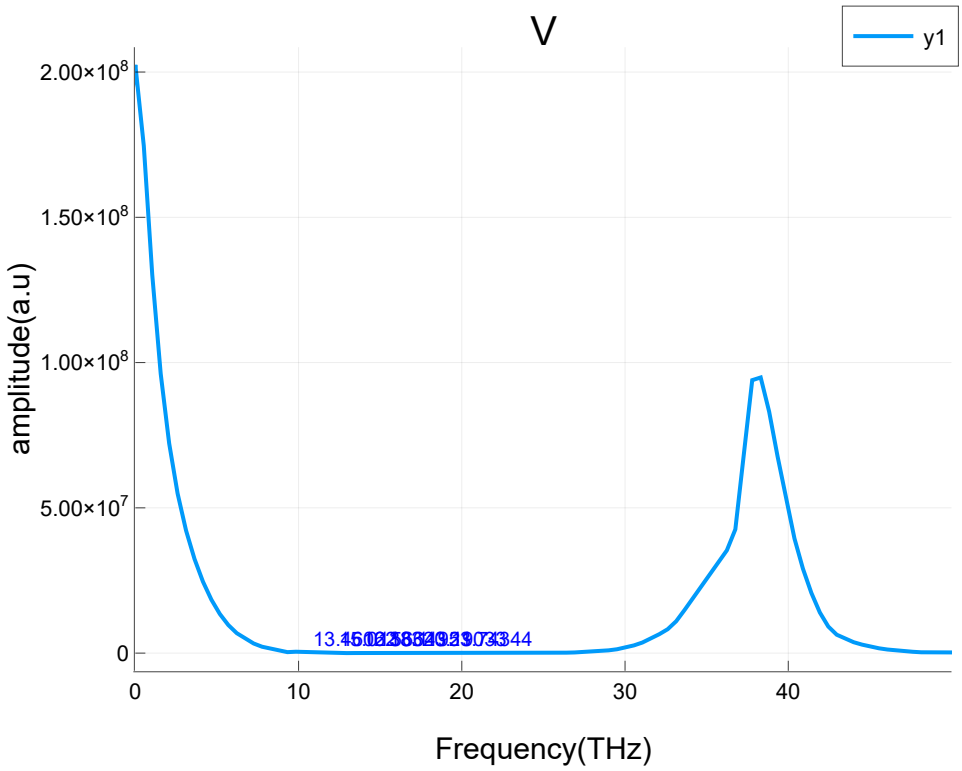


IR Mode


Q₂ (Ω₂=18.60383 THz)
18.63723
2.00×10^5
1.50×10^5
1.00×10^5
5.00×10^4
0
amplitude(a.u)
6.21247
7.76551
9.83632
11.90712
13.46022
23.81424
0
5
10
15
20
Frequency(THz)


Q1+Q2
y1
18.63723
2.00×10^5
1.50×10^5
1.00×10^5
5.00×10^4
0
4.65931
0.0
7.24781
9.31862
11.90712
13.46022
23.81424
0
10
20

Potential

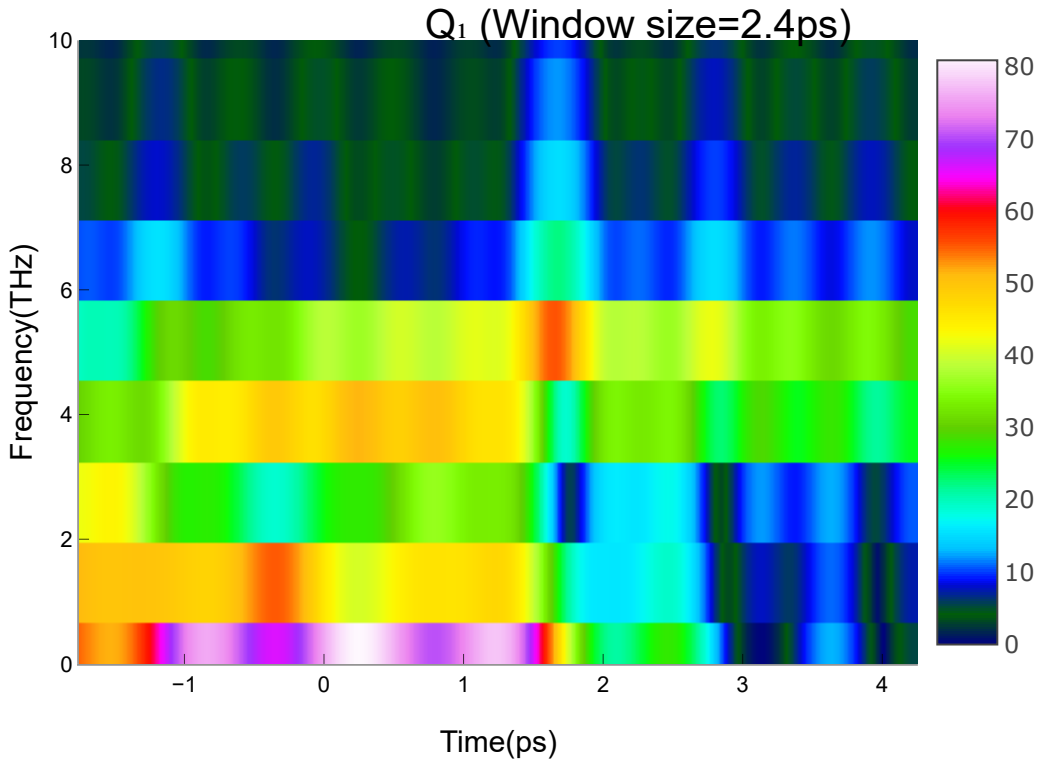

V
y1
2.00×10^8
1.50×10^8
1.00×10^8
5.00×10^7
0
amplitude(a.u)
0
10
20
30
40
Frequency(THz)

Short-time FT

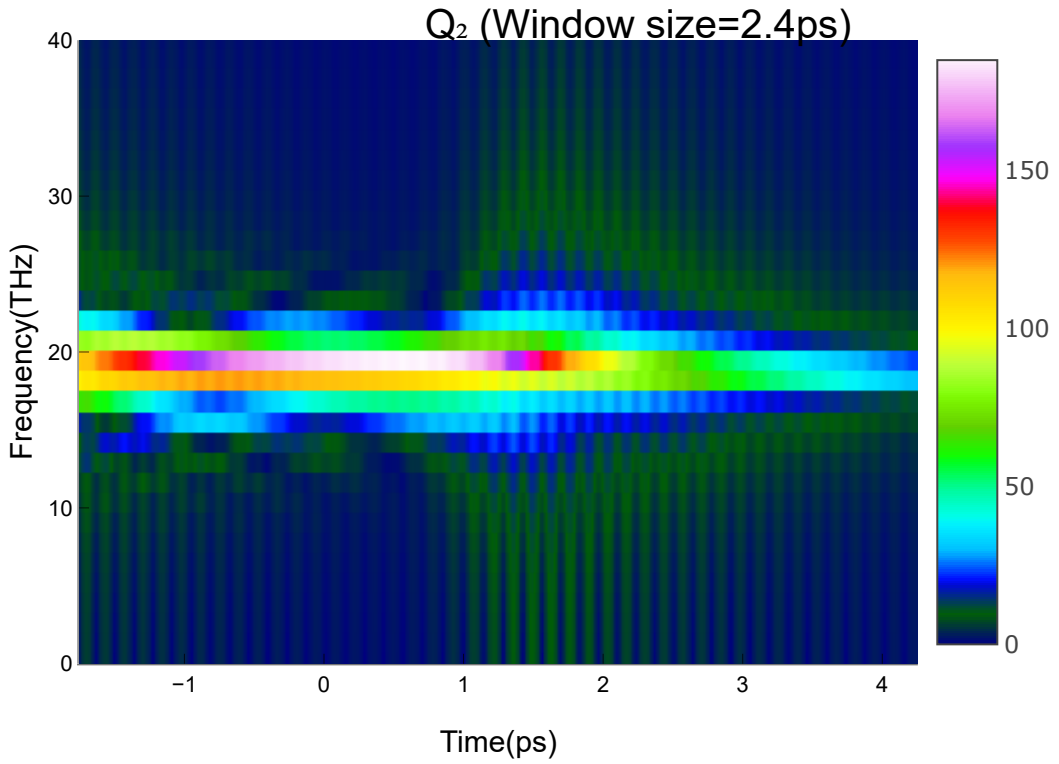
Window size percentage: 0.4

```
[0.0, 1.29296, 2.58592, 3.87888, 5.17184, 6.4648, 7.75776, 9.05071, 10.3437, 11.6366, 12.9
```

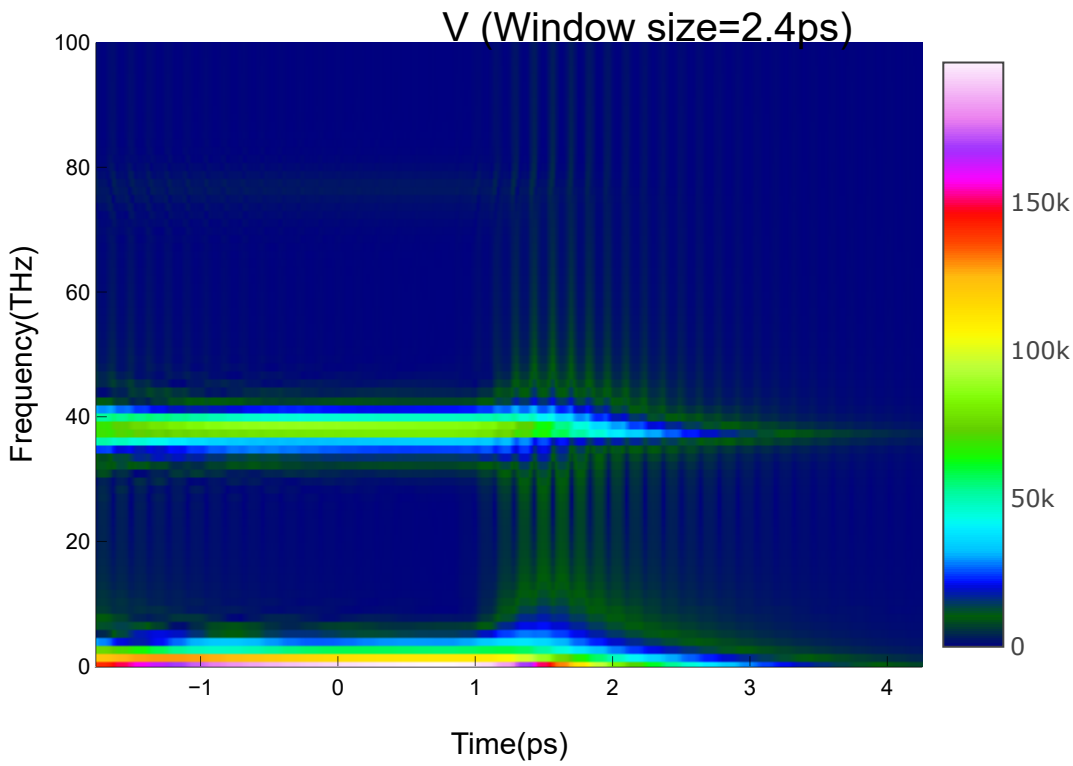
Raman Mode



IR Mode



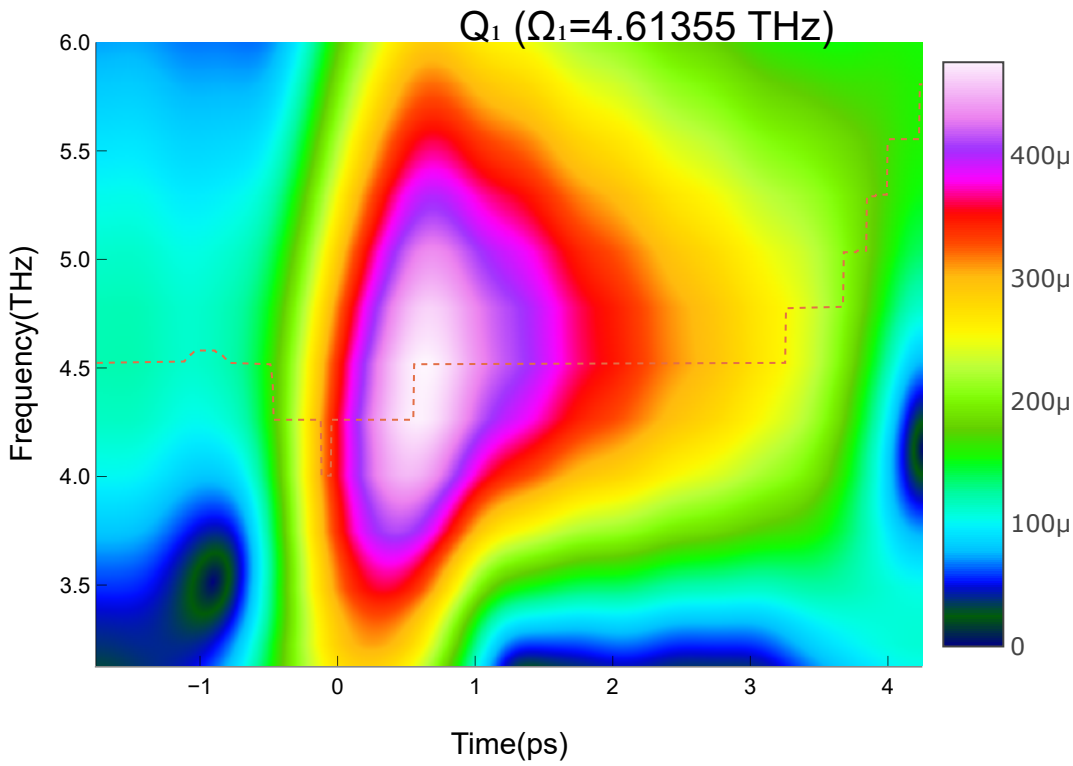
Potential



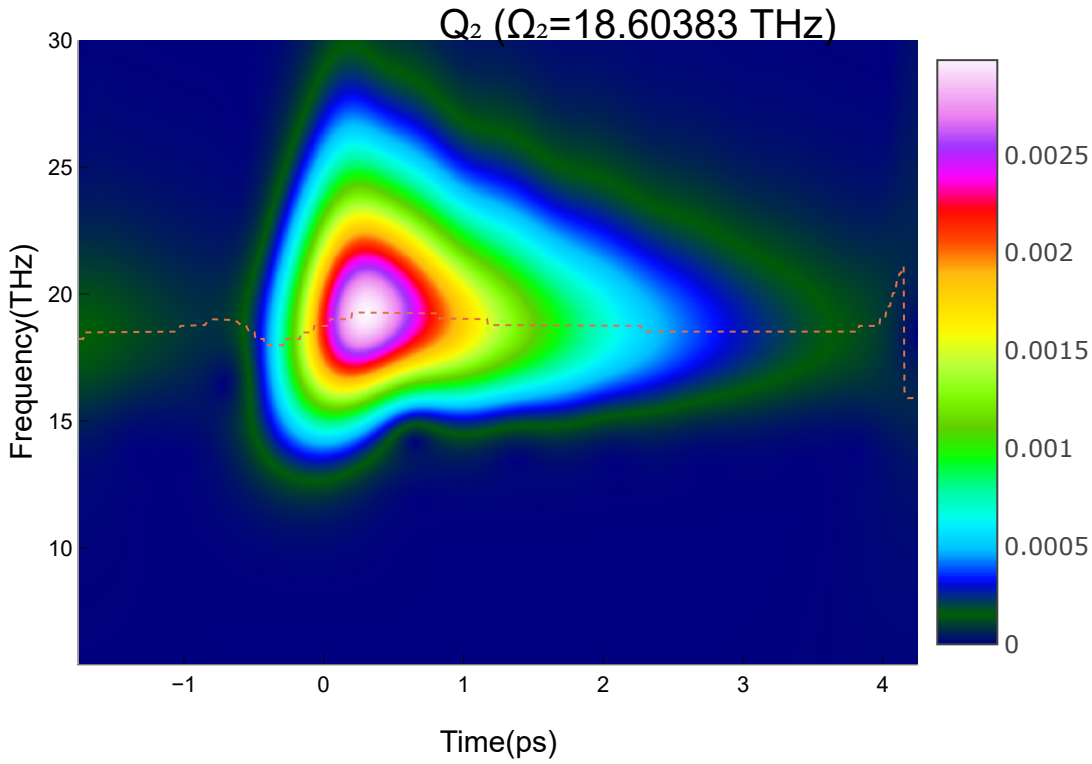
Wavelet Transforms

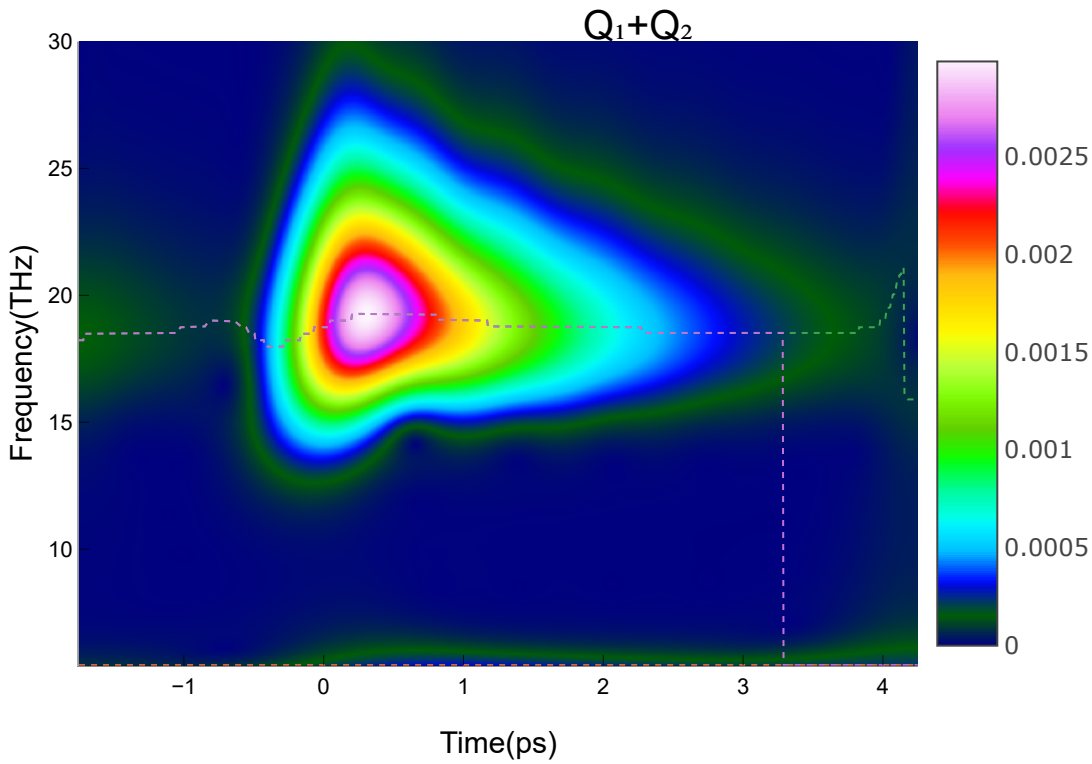
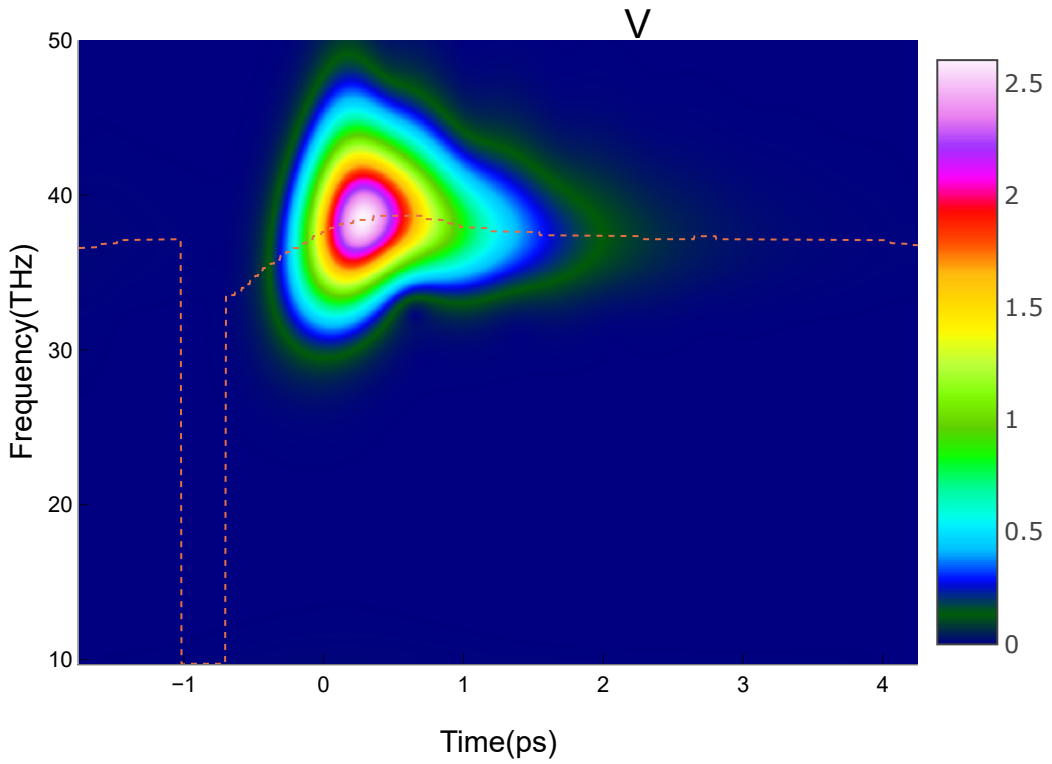
```
Wt (generic function with 5 methods)
```

Raman Mode



IR Mode


Q2 (Ω2=18.60383 THz)
Frequency(THz)
Time(ps)
30
25
20
15
10
−1
0
1
2
3
4
0.0025
0.002
0.0015
0.001
0.0005
0


Q1+Q2
Frequency(THz)
Time(ps)
30
25
20
15
10
−1
0
1
2
3
4
0.0025
0.002
0.0015
0.001
0.0005
0

Potential